Flight Control I

Equilibrium, Linearisation, PD/PID Control

and Motor Dynamics

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What you should be able to do

Find the state and thrust at stationary hover.

Linearise a nonlinear model near an operating point.

Choose PD gains from damping and natural frequency.

Explain when integral action helps.

Explain why motor lag changes closed-loop behaviour.

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Coordinates, signs and symbols

Use an inertial vertical coordinate z, positive downwards.

$T \geq 0$ is the magnitude of total upward thrust.

$w = \dot{z}$ for level vertical motion with no body rotation.

$e = z^d - z$. A climb has a more negative z.

u_z is a commanded acceleration in m/s², not thrust.

Use radians in equations. Take g = 9.81 m/s².

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From forces to a vertical model

Assume a level aircraft, constant mass and no drag.

$\Sigma F_z = mg - T = m\ddot{z}$

$\dot{z} = w$

$\dot{w} = g - T/m$

Position and velocity are the two states.

The physical input is total thrust T in newtons.

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Stationary hover is an equilibrium

At equilibrium the state does not change.

$w_0 = 0$ and $\dot{w}_0 = 0$

$T_0 = mg$

The chosen position z_0 can be any constant.

T = mg alone gives zero acceleration.

An existing vertical velocity would remain unchanged.

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Practice: find the hover point

A level UAV has mass m = 1.50 kg.

1. Find T_0 for stationary hover.

2. If $T = T_0$ but w = +0.40 m/s, is it hovering?

3. For upward acceleration 1.00 m/s², find T.

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Solution: hover and acceleration

$T_0 = 1.50(9.81) = 14.715\ \text{N} \approx 14.7\ \text{N}.$

$w = +0.40$ m/s means descent, so this is not hover.

Upward acceleration means $\ddot{z} = -1.00\ \text{m/s}^2$.

$T = m(g - \ddot{z})$

$T = 1.50[9.81 - (-1.00)] = 16.215\ \text{N} \approx 16.2\ \text{N}.$

Thrust exceeds weight to accelerate upwards.

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Why linearise?

Rotor thrust contains Ω^2. Tilt introduces sin θ and cos θ.

Linear controllers use a local approximation near a chosen state.

Write $x = x_0 + \delta x$ and $u = u_0 + \delta u$.

δ means a small deviation from the operating point.

The approximation is local, not a replacement for all flight.

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Reading the linearisation notation

x: state vector u: input vector f: state equations

x_0, u_0: equilibrium values, with $f(x_0,u_0) = 0$

$\delta x = x - x_0$ $\delta u = u - u_0$

A: sensitivity to states B: sensitivity to inputs

∂ means partial derivative: vary one quantity at a time.

δ means deviation. A dot means time derivative.

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Example: linearise rotor thrust

$T_i = k_T \Omega_i^2$ and $\Omega_i = \Omega_0 + \delta\Omega_i$

$T_i = k_T \Omega_0^2 + 2k_T \Omega_0 \delta\Omega_i + k_T(\delta\Omega_i)^2$

$\delta T_i \approx 2k_T \Omega_0 \delta\Omega_i$

At $\Omega_0 = 500$ rad/s and $k_T = 2.00 \times 10^{-5}$:

$\delta\Omega_i = 10$ rad/s gives $\delta T_i \approx 0.200$ N.

The exact increase is 0.202 N.

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Example: linearise the planar model

Positive pitch θ is nose-up. Thrust tilts towards −x.

$\ddot{x} = -(T/m) \sin\theta$

$\ddot{z} = g - (T/m) \cos\theta \qquad \ddot{\theta} = M_y/J_y$

About hover: $\theta_0 = 0$, $T_0 = mg$, $M_{y0} = 0$.

$\delta\ddot{x} \approx -g\,\delta\theta \qquad \delta\ddot{z} \approx -\delta T/m$

$\delta\ddot{\theta} = \delta M_y/J_y$

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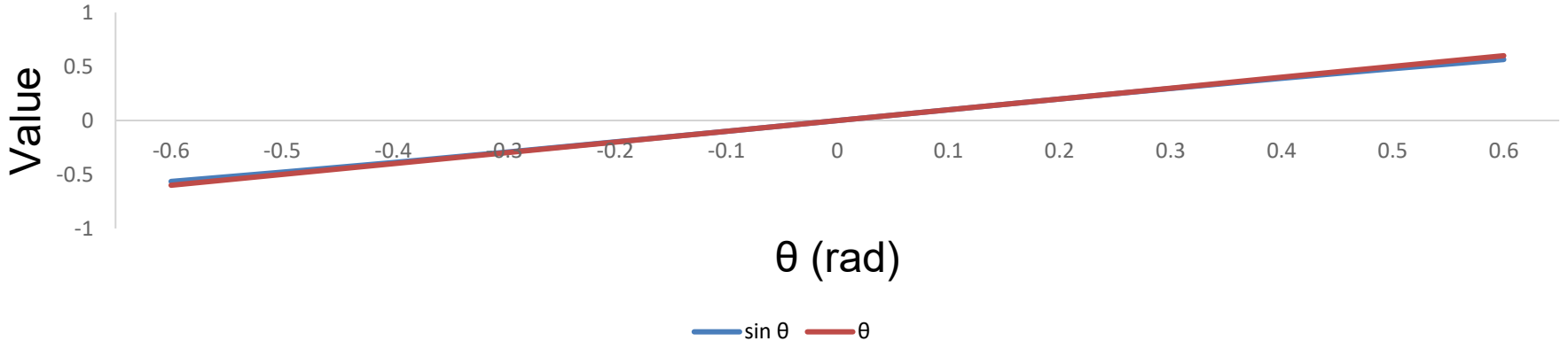
How small is a small angle?

sin θ ≈ θ and cos θ ≈ 1, with θ in radians.

At 5°: θ = 0.0873, sin θ = 0.0872.

At 20°: θ = 0.349, sin θ = 0.342, cos θ = 0.940.

Near level hover, the small-angle model is accurate. With larger tilt, we must account for the reduction in vertical thrust (Larger tilt changes the vertical thrust component).



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Acceleration command and the plant

Choose u_z as the desired vertical acceleration.

$T_{cmd} = m(g - u_z)$

With ideal thrust tracking: $\ddot{z} = u_z$. (Actual thrust = commanded thrust)

With zero initial conditions: $Z(s)/U_z(s) = 1/s^2$.

Two integrations take acceleration to velocity to position.

We let the controller request acceleration, then convert that request into thrust. If thrust follows the command perfectly, the UAV behaves as a double integrator: acceleration integrates into velocity, and velocity integrates into position. We will now design PD control for this model.

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PD control: correct error and damp motion

$e = z^d - z$

$u_z = k_p\, e + k_d\, \dot{e}$

For a constant reference: $\dot{e} = -\dot{z}$.

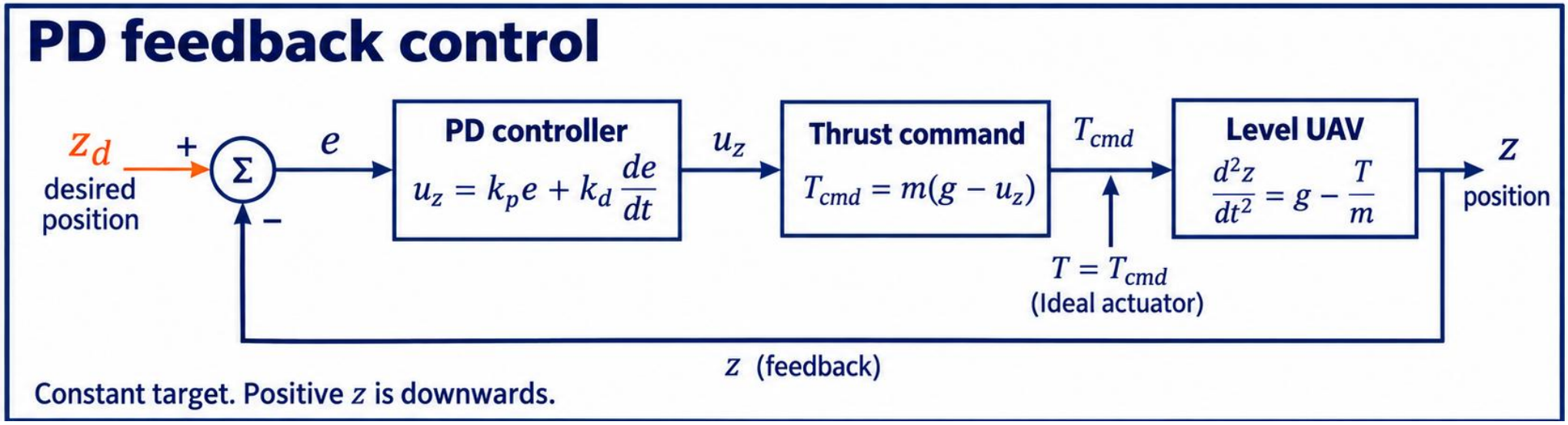
$k_p\, e$ accelerates towards the desired position.

$k_d\, \dot{e}$ opposes unwanted velocity.

Gain units: k_p in s^{-2} and k_d in s^{-1}.

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The complete PD feedback loop



The output returns to the negative input: $e = z^d - z$.

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PD symbols and the desired response

z^d: desired position [m] z: actual position [m]

$e = z^d - z$ [m] $\dot{e} = \dot{z}^d - \dot{z}$ [m/s]

u_z: acceleration command [m/s²]

k_p: proportional gain [s⁻²] k_d: derivative gain [s⁻¹]

ω_n: natural frequency [rad/s] ζ: damping ratio [no units]

$k_p = \omega_n^2$ $k_d = 2\zeta\omega_n$

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Derive the error equation

For a constant reference: $\ddot{e} = -\ddot{z}$.

Substitute $\ddot{z} = u_z = k_p e + k_d \dot{e}$.

$\ddot{e} + k_d \dot{e} + k_p e = 0$

Characteristic polynomial: $s^2 + k_d\, s + k_p$

Compare with $s^2 + 2\zeta\omega_n s + \omega_n^2$.

$k_p = \omega_n^2 \qquad k_d = 2\zeta\omega_n$

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Worked PD design

Choose $\omega_n = 4.00$ rad/s and $\zeta = 0.750$.

$k_p = 4^2 = 16.0\ s^{-2}$

$k_d = 2(0.750)(4) = 6.00\ s^{-1}$

$s^2 + 6s + 16 = 0$

Poles: $s = -3.00 \pm j2.65\ s^{-1}$.

Negative real parts mean the error decays.
The imaginary part means the response can oscillate.
Together, they describe a **decaying oscillatory response**.

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Practice: command a climb

Use m = 1.50 kg, k_p = 16.0 and k_d = 6.00.

At this instant: z^d = −1.00 m, z = −0.80 m.

The reference is constant and ż = +0.10 m/s.

Find e, ė, u_z and T_{cmd}.

Explain the direction of acceleration and compare T_{cmd} with mg.

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Solution: command a climb

$e = -1.00 - (-0.80) = -0.200$ m.

$\dot{e} = 0 - 0.10 = -0.100$ m/s.

$u_z = 16(-0.200) + 6(-0.100) = -3.80$ m/s².

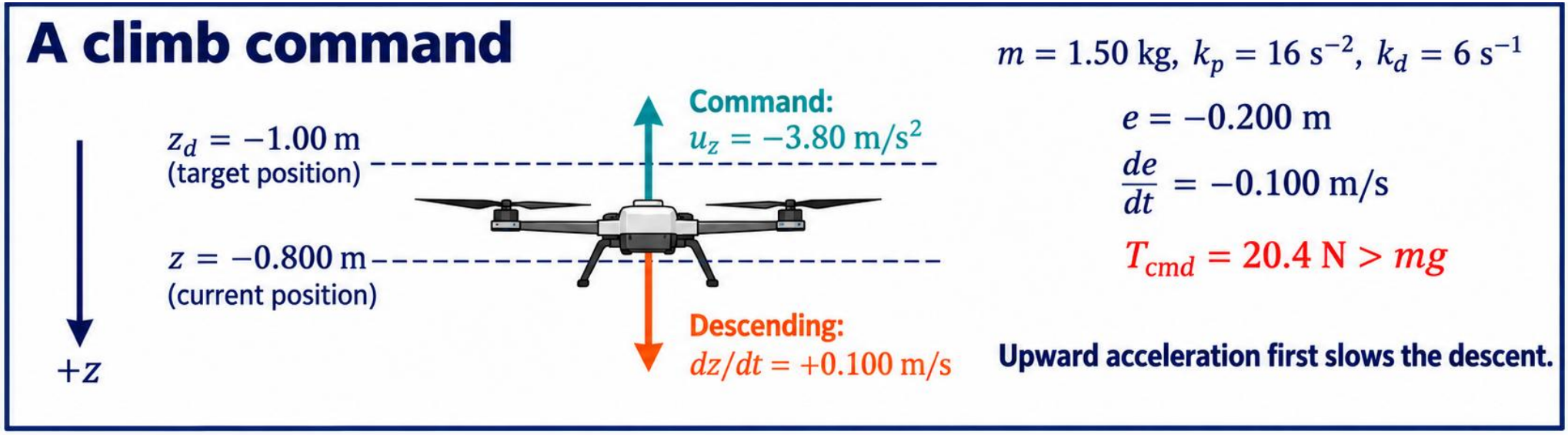
$T_{cmd} = 1.50[9.81 - (-3.80)] = 20.415$ N.

$T_{cmd} \approx 20.4$ N, above the 14.7 N hover value.

Upward acceleration initially slows the descent.

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The signs in a climb command



Positive velocity: descending. Negative acceleration: slowing that descent.

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Tracking a moving reference

For a smooth trajectory, $\ddot{z}^d$ need not be zero.

$u_z = \ddot{z}^d + k_p(z^d - z) + k_d(\dot{z}^d - \dot{z})$

Then $\ddot{e} = \ddot{z}^d - \ddot{z} = -k_p e - k_d \dot{e}$.

Feedforward $\ddot{z}^d$ supplies the planned acceleration.

Feedback corrects deviations from the plan.

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What the poles tell us

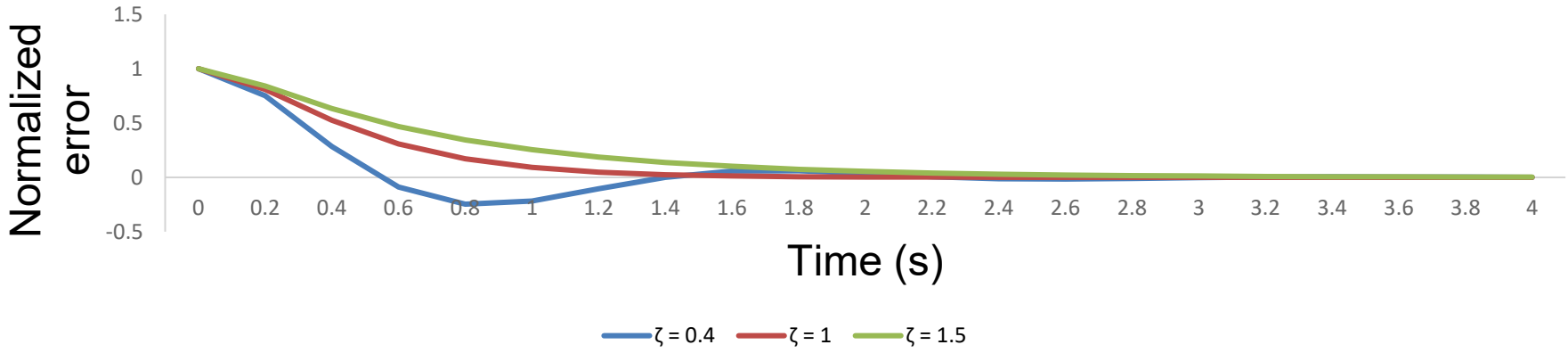
$k_p > 0$ and $k_d > 0$ stabilise the ideal double integrator.

$0 < \zeta < 1$: oscillatory decay.

$\zeta = 1$: critical damping. $\zeta > 1$: overdamped.

Approximate 2% settling time: $t_s \approx 4/(\zeta\omega_n)$.

The source's $4.6/(\zeta\omega_n)$ corresponds to a 1% convention.



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Why PD can leave an offset

Suppose actual acceleration is $\ddot{z} = u_z + d_0$.

d_0 is a constant unmodelled acceleration.

With PD: $\ddot{e} + k_d\dot{e} + k_p e = -d_0$.

At steady state: $e_{ss} = -d_0/k_p$.

Example: $d_0 = +0.80$ m/s², $k_p = 16$ s⁻².

$e_{ss} = -0.050$ m. The aircraft sits 5 cm too low.

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PID adds a memory of error

$\dot{\eta} = e \qquad \eta(t) = \int e \, dt$

$u_z = k_p e + k_d \dot{e} + k_i \eta$

Persistent error accumulates in η.

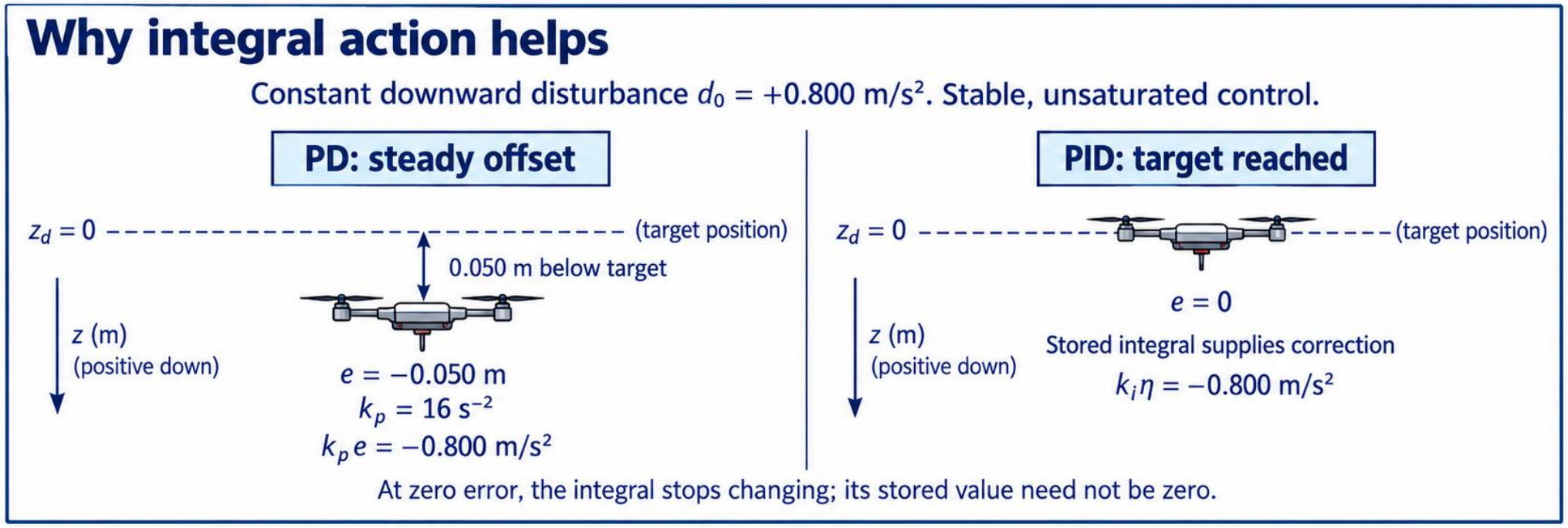
At a stable unsaturated steady state, $\dot{\eta} = e = 0$.

The integral term can supply $u_z = -d_0$.

For acceleration control, k_i has units s^{-3}.

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PD offset and integral correction



η stores accumulated error [m s]. Its value can remain nonzero when e = 0.

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PID changes the characteristic polynomial

For constant reference and no actuator lag:

$s^3 + k_d s^2 + k_p s + k_i = 0$

Positive gains alone are not sufficient.

For this cubic, stability requires $k_d k_p > k_i > 0$.

With $k_p = 16$ and $k_d = 6$, choose k_i below 96 s^{-3}.

This is a stability bound, not an optimum tuning rule.

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Practical PD/PID implementation

Thrust is bounded: $0 \leq T \leq T_{max}$.

A saturated actuator cannot follow a larger command.

Integral windup: error continues accumulating during saturation.

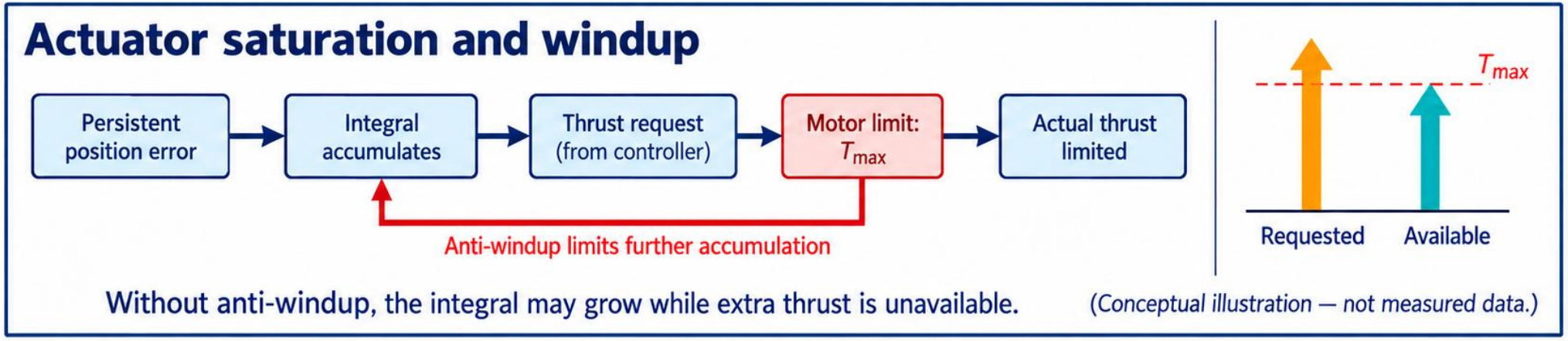
Use conditional integration or back-calculation.

Derivative action amplifies noise. Use a filtered velocity estimate.

For setpoint steps, derivative on measurement avoids derivative kick.

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Saturation and integral windup

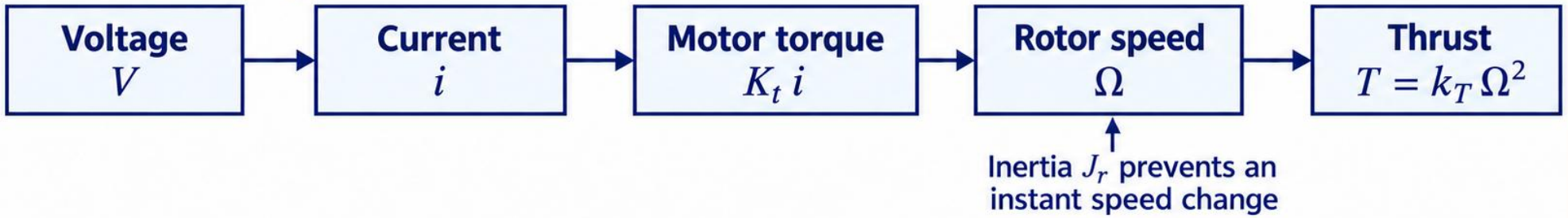


Anti-windup prevents integration from driving the actuator further into saturation.

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Motor and propeller physics

From voltage to thrust



$$L_a \frac{di}{dt} + R_a i + K_e \Omega = V \qquad J_r \frac{d\Omega}{dt} = K_t i - B_m \Omega - Q_{\text{load}}$$

Conceptual chain; back EMF and load torques appear in the equations.

Voltage changes current. Net torque changes speed. Speed sets thrust.

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Electrical equation: every symbol

$L_a \, di/dt + R_a \, i + K_e \, \Omega = V$

V: applied voltage [V] i: motor current [A]

L_a: equivalent winding inductance [H]

R_a: equivalent winding resistance [Ω, ohms]

K_e: back-EMF constant [V/(rad/s)]

Ω: rotor angular speed [rad/s]

$L_a \, di/dt$, $R_a \, i$ and $K_e \, \Omega$ are all voltage terms.

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Mechanical equation: every symbol

$J_r \, d\Omega/dt = K_t \, i - B_m \, \Omega - Q_{load}$

J_r: effective rotating inertia [kg m²]

K_t: motor torque constant [N m/A]

B_m: viscous damping coefficient [N m/(rad/s)]

Q_{load}: resisting propeller torque [N m]

Net torque = inertia × angular acceleration.

K_t (motor torque) differs from k_T (propeller thrust).

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Linearise the motor load

Near Ω_0: $\delta Q_{load} \approx K_Q \, \delta\Omega$.

For $Q_{load} = k_M \, \Omega^2$: $K_Q = 2k_M \, \Omega_0$.

If L_a/R_a is much shorter than the mechanical time scale:

$\delta i \approx (\delta V - K_e \, \delta\Omega)/R_a$.

$J_r \, \delta\dot{\Omega} + B_{eff} \, \delta\Omega = (K_t/R_a) \, \delta V$.

$B_{eff} = B_m + K_Q + K_t \, K_e/R_a$.

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Motor linearisation: coefficients and units

Ω_0: steady operating speed [rad/s]

k_M: propeller drag-torque coefficient [N m/(rad/s)²]

$K_Q = 2k_M \Omega_0$: local load slope [N m/(rad/s)]

$B_{eff} = B_m + K_Q + K_t K_e/R_a$

B_{eff}: effective damping [N m/(rad/s)]

L_a/R_a: electrical time constant [s]

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Time constant, gain and thrust command

$\tau_m = J_r/B_{eff}$: mechanical time constant [s]

$K_\Omega = K_t/(R_a B_{eff})$: speed gain [(rad/s)/V]

$\delta\Omega(s)/\delta V(s) = K_\Omega/(\tau_m s + 1)$

Unit steady-state gain means actual change reaches its command.

$\tau_m \delta\dot{T} + \delta T = \delta T_{cmd}$ is a calibrated local model.

$\delta\dot{T} = (\delta T_{cmd} - \delta T)/\tau_m$: thrust catches up gradually.

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How fast does a motor respond?

For a step command and zero initial deviation:

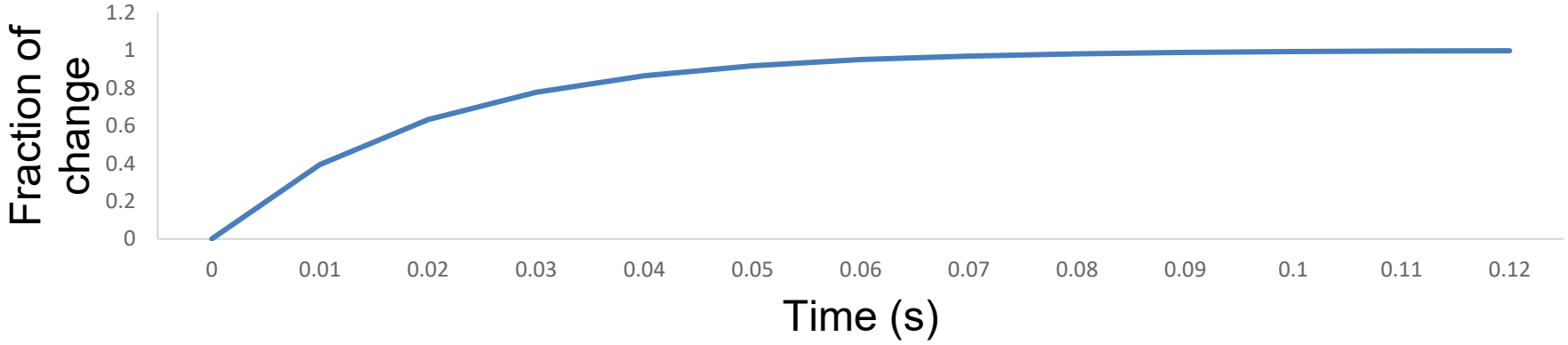
$\delta\Omega(t)/\delta\Omega_{cmd} = 1 - \exp(-t/\tau_m)$.

At τ_m: 63.2% of the change.

At $3\tau_m$: 95.0%. At $5\tau_m$: 99.3%.

If $\tau_m = 0.020$ s, $3\tau_m = 0.060$ s.

$1/\tau_m = 50\ s^{-1}$ is a pole rate, not a second-order ω_n.



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Motor lag adds a closed-loop pole

With a normalized actuator lag: $G(s) = 1/[s^2(\tau_m s + 1)]$.

PD control gives $\tau_m s^3 + s^2 + k_d s + k_p = 0$.

For positive τ_m, k_p and k_d, stability requires:

$k_d > \tau_m k_p$

A design stable with an ideal actuator may fail with motor lag.

Check the complete model before increasing gains.

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Practice: test the actuator-aware design

Use $k_p = 64.0\ s^{-2}$ and $k_d = 11.2\ s^{-1}$.

1. Check stability for $\tau_m = 0.020$ s.

2. Repeat for $\tau_m = 0.200$ s.

3. Does passing the inequality guarantee good tracking?

Use $k_d > \tau_m k_p$.

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Solution: motor lag matters

For 0.020 s: $\tau_m k_p = 1.28\ s^{-1} < 11.2\ s^{-1}$.

The simplified closed loop is stable.

For 0.200 s: $\tau_m k_p = 12.8\ s^{-1} > 11.2\ s^{-1}$.

The simplified closed loop is unstable.

Stability does not guarantee acceptable overshoot or settling.

Also check saturation, noise and neglected dynamics.

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Motor response and closed-loop stability

Motor lag and control

First-order response to a step

Elapsed time	Change achieved
0	0%
τ_m	63.2%
$3\tau_m$	95.0%
$5\tau_m$	99.3%

Fraction of the requested change

PD with first-order actuator lag

For positive parameters: $k_d > \tau_m k_p$

$\tau_m = 0.020$ s: $11.2 > 1.28$	Stable in this model
$\tau_m = 0.200$ s: $11.2 < 12.8$	Unstable in this model

A slower actuator can destabilise the same PD controller.

Example gains: $k_p = 64.0\ s^{-2}$ and $k_d = 11.2\ s^{-1}$.

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From equilibrium to a controller

1. Define coordinates and the physical input.
2. Find an equilibrium and define deviations.
3. Linearise only where necessary.
4. Choose gains and check the error dynamics.
5. Include actuator dynamics and limits.

Next lecture: connect position control to attitude control.

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